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CHEMISTRY
HIGHER LEVEL
PAPER 1

SPECIMEN PAPER

1 hour

INSTRUCTIONS TO CANDIDATES

- Do not open this examination paper until instructed to do so.
- Answer all the questions.
- For each question, choose the answer you consider to be the best and indicate your choice on the answer sheet provided.
- The periodic table is provided for reference on page 2 of this examination paper.
- The maximum mark for this examination paper is *[40 marks]*.

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The Periodic Table

Atomic number

Element

Relative atomic mass

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 **H** 1.01																	2 **He** 4.00
2	3 **Li** 6.94	4 **Be** 9.01											5 **B** 10.81	6 **C** 12.01	7 **N** 14.01	8 **O** 16.00	9 **F** 19.00	10 **Ne** 20.18
3	11 **Na** 22.99	12 **Mg** 24.31											13 **Al** 26.98	14 **Si** 28.09	15 **P** 30.97	16 **S** 32.07	17 **Cl** 35.45	18 **Ar** 39.95
4	19 **K** 39.10	20 **Ca** 40.08	21 **Sc** 44.96	22 **Ti** 47.87	23 **V** 50.94	24 **Cr** 52.00	25 **Mn** 54.94	26 **Fe** 55.85	27 **Co** 58.93	28 **Ni** 58.69	29 **Cu** 63.55	30 **Zn** 65.38	31 **Ga** 69.72	32 **Ge** 72.63	33 **As** 74.92	34 **Se** 78.96	35 **Br** 79.90	36 **Kr** 83.90
5	37 **Rb** 85.47	38 **Sr** 87.62	39 **Y** 88.91	40 **Zr** 91.22	41 **Nb** 92.91	42 **Mo** 95.96	43 **Tc** (98)	44 **Ru** 101.07	45 **Rh** 102.91	46 **Pd** 106.42	47 **Ag** 107.87	48 **Cd** 112.41	49 **In** 114.82	50 **Sn** 118.71	51 **Sb** 121.76	52 **Te** 127.60	53 **I** 126.90	54 **Xe** 131.29
6	55 **Cs** 132.91	56 **Ba** 137.33	57 † **La** 138.91	72 **Hf** 178.49	73 **Ta** 180.95	74 **W** 183.84	75 **Re** 186.21	76 **Os** 190.23	77 **Ir** 192.22	78 **Pt** 195.08	79 **Au** 196.97	80 **Hg** 200.59	81 **Tl** 204.38	82 **Pb** 207.2	83 **Bi** 208.98	84 **Po** (209)	85 **At** (210)	86 **Rn** (222)
7	87 **Fr** (223)	88 **Ra** (226)	89 ‡ **Ac** (227)	104 **Rf** (267)	105 **Db** (268)	106 **Sg** (269)	107 **Bh** (270)	108 **Hs** (269)	109 **Mt** (278)	110 **Ds** (281)	111 **Rg** (281)	112 **Cn** (285)	113 **Unt** (286)	114 **Uug** (289)	115 **Uup** (288)	116 **Uuh** (293)	117 **Uus** (294)	118 **Uuo** (294)

†	58 **Ce** 140.12	59 **Pr** 140.91	60 **Nd** 144.24	61 **Pm** (145)	62 **Sm** 150.36	63 **Eu** 151.96	64 **Gd** 157.25	65 **Tb** 158.93	66 **Dy** 162.50	67 **Ho** 164.93	68 **Er** 167.26	69 **Tm** 168.93	70 **Yb** 173.05	71 **Lu** 174.97
‡	90 **Th** 232.04	91 **Pa** 231.04	92 **U** 238.03	93 **Np** (237)	94 **Pu** (244)	95 **Am** (243)	96 **Cm** (247)	97 **Bk** (247)	98 **Cf** (251)	99 **Es** (252)	100 **Fm** (257)	101 **Md** (258)	102 **No** (259)	103 **Lr** (262)

1. Which changes of state are endothermic processes?

I. Condensing

II. Melting

III. Subliming

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

2. What is the sum of the coefficients when the equation for the combustion of ammonia is balanced using the smallest possible whole numbers?

$$___\ NH_3(g) + ___\ O_2(g) \rightarrow ___\ N_2(g) + ___\ H_2O(g)$$

A. 6

B. 12

C. 14

D. 15

3. 5.00 g of calcium carbonate, when heated, produced 2.40 g of calcium oxide. Which is the correct expression for the percentage yield of calcium oxide? ($M_r(CaCO_3) = 100$; $M_r(CaO) = 56$.)

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

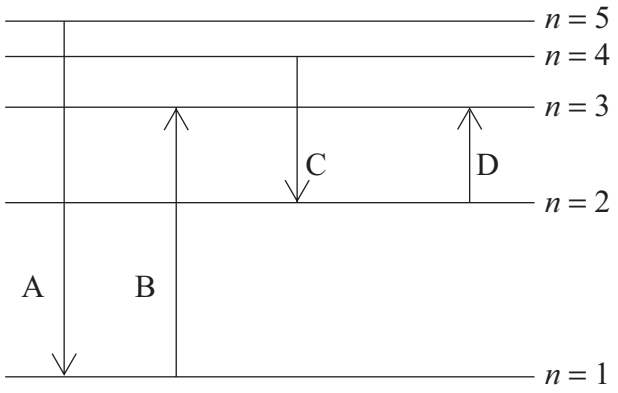
A. $\frac{56 \times 5.00 \times 100}{2.40}$

B. $\frac{2.40 \times 100 \times 100}{56 \times 5.00}$

C. $\frac{56 \times 5.00 \times 100}{2.40 \times 100}$

D. $\frac{2.40 \times 100}{56 \times 5.00}$

4. Which electronic transition would absorb the radiation of the shortest wavelength?



5. Which is the electron configuration of the ion Fe^{2+}?

A. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$

B. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6\ 4s^2$

C. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^4\ 4s^2$

D. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$

6. Which element is in group 2?

	1st ionization energy / kJ mol^{-1}	**2nd ionization energy / kJ mol^{-1}**	**3rd ionization energy / kJ mol^{-1}**	**4th ionization energy / kJ mol^{-1}**
A.	1402	2856	4578	7475
B.	590	1145	4912	6474
C.	403	2632	3900	5080
D.	578	1817	2745	11578

7. Which element is in the f-block of the periodic table?

A. Be

B. Ce

C. Ge

D. Re

8. Which property increases down group 1 of the periodic table?

A. Melting point

B. First ionization energy

C. Atomic radius

D. Electronegativity

9. What is the overall charge on the complex ion formed by iron(II) and six cyanide ions, CN^-?

A. 4+

B. 4–

C. 8–

D. 8+

10. Which statement about transition metal complex ions is correct?

A. The difference in energy of the d orbitals is independent of the oxidation state of the metal.

B. The colour of the complex is caused by light emitted when an electron falls back from a higher to a lower energy level.

C. The colour of the complex is the colour of the light absorbed when an electron moves from a lower to a higher energy level.

D. The difference in energy of the d orbitals depends on the nature of the ligand.

11. Which is the best description of ionic bonding?

A. Electrostatic attraction between oppositely charged ions

B. Electrostatic attraction between positive ions and electrons

C. Electrostatic attraction of nuclei towards shared electrons in the bond between the nuclei

D. Electrostatic attraction between nuclei

12. Which intermolecular forces are covered by the term van der Waals'?

I. London dispersion forces

II. Dipole-induced dipole forces

III. Dipole-dipole forces

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

13. Which bond is the **least** polar?

A. C=O in CO_2

B. C–H in CH_4

C. C–Cl in CCl_4

D. N–H in CH_3NH_2

14. Which pair of compounds contains 9 sigma, σ, and 2 pi, π, bonds in each molecule?

A. CH_3CO_2H and $CH_3CH(OH)CH_3$

B. CH_3COCH_3 and $CH_3COOCH_2CH_3$

C. $CHCCH_2CH_3$ and $CH_2CHCHCH_2$

D. CH_3COH and CH_3CH_2OH

15. Which molecule contains an atom with sp^2 hybridization?

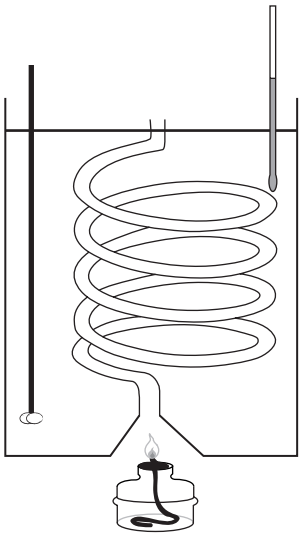
A. $CH_3CH_2CH_2NH_2$

B. $CH_3CH_2CH_2CN$

C. $CH_3CH_2CH_2CH_2Cl$

D. $CH_3CH_2CHCHCH_3$

16. When 0.46 g of ethanol is burned under a water-filled calorimeter, the temperature of 500 g of water is raised by 3.0 K. (Molar mass of ethanol = 46 g mol^{-1}; specific heat capacity of water = 4.18 J g^{-1} K^{-1}; $q = mc\Delta T$.)



What is the expression for the enthalpy of combustion, ΔH_c, in kJ mol^{-1}?

A. $-\dfrac{500 \times 4.18 \times 3.0 \times 46}{0.46}$

B. $-\dfrac{500 \times 4.18 \times (273 + 3.0) \times 46}{0.46 \times 1000}$

C. $-\dfrac{500 \times 4.18 \times 3.0 \times 46}{0.46 \times 1000}$

D. $-\dfrac{0.46 \times 1000}{500 \times 4.18 \times 3.0 \times 46}$

17. Given the following information, what is the standard enthalpy of formation, $\Delta H^{\ominus}_{f}$, of methane?

$$C(s) + O_2(g) \rightarrow CO_2(g) \qquad \Delta H = E \text{ kJ}$$
$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(l) \qquad \Delta H = F \text{ kJ}$$
$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) \qquad \Delta H = G \text{ kJ}$$

A. E + F + G

B. E + F – G

C. E + 2F + G

D. E + 2F – G

18. Which combination has the most endothermic lattice enthalpy?

	Radius of positive ion / nm	**Radius of negative ion / nm**	**Charge on positive ion**	**Charge on negative ion**
A.	0.100	0.185	2+	2–
B.	0.102	0.180	1+	1–
C.	0.149	0.180	1+	1–
D.	0.100	0.140	2+	2–

19. In which reaction is the value of ΔS positive?

A. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

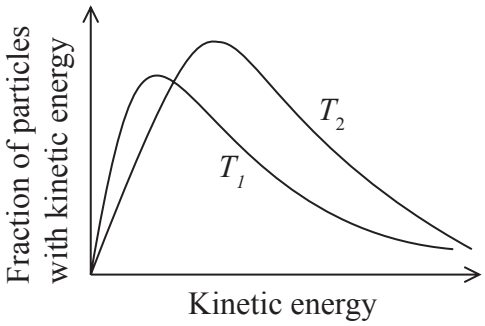
B. $H_2O(g) \rightarrow H_2O(s)$

C. $2KI(aq) + Pb(NO_3)_2(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$

D. $2ZnS(s) + 3O_2(g) \rightarrow 2ZnO(s) + 2SO_2(g)$

20. Which graph shows the Maxwell-Boltzmann energy distribution of a same amount of a gas at two temperatures, where T_2 is greater than T_1?

A.



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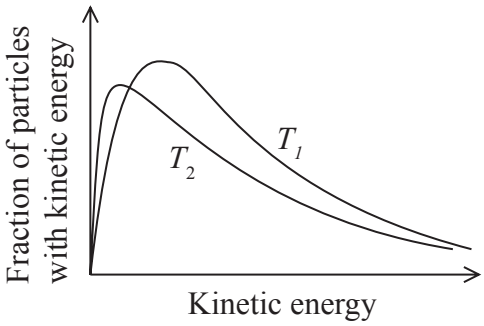
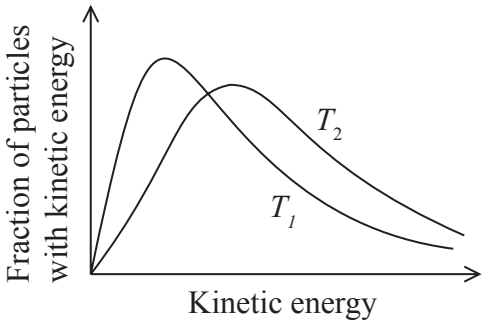
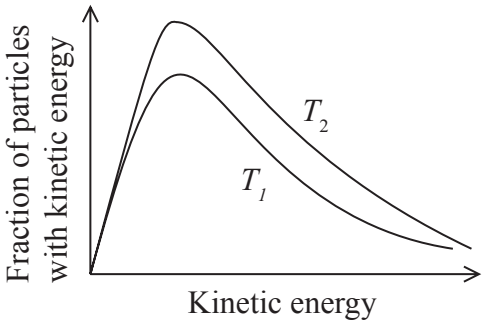


C.



D.



21. Which changes increase the rate of this reaction, other conditions remaining constant?

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

I. Using larger lumps of calcium carbonate

II. Increasing the temperature of the reaction mixture

III. Increasing the concentration of hydrochloric acid

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

22. The rate information below was obtained for the following reaction at a constant temperature.

$$H_2O_2(aq) + 2H^+(aq) + 2I^-(aq) \rightarrow 2H_2O(l) + I_2(aq)$$

Initial [H_2O_2 (aq)] / mol dm^{-3}	**Initial [H^+(aq)] / mol dm^{-3}**	**Initial [I^-(aq)] / mol dm^{-3}**	**Initial rate of reaction / mol dm^{-3} s^{-1}**
0.005	0.05	0.015	1.31×10^{-6}
0.01	0.05	0.015	2.63×10^{-6}
0.01	0.05	0.03	5.25×10^{-6}
0.01	0.1	0.03	5.25×10^{-6}

What is the overall order of the reaction?

A. 0

B. 1

C. 2

D. 3

23. Which reaction is **most** likely to be spontaneous?

	Enthalpy change	**Entropy**
A.	exothermic	entropy decreases
B.	exothermic	entropy increases
C.	endothermic	entropy decreases
D.	endothermic	entropy increases

24. Which conditions give the greatest equilibrium yield of methanal, $H_2CO(g)$?

$$CO(g) + H_2(g) \rightleftharpoons H_2CO(g) \qquad \Delta H = -1.8\,kJ$$

	Pressure	Temperature
A.	high	low
B.	high	high
C.	low	high
D.	low	low

25. Which combination of temperature and equilibrium constant is most typical of a reaction going to completion? (Refer to the equation $\Delta G = -RT \ln K$.)

	Temperature	Equilibrium constant
A.	high	> 1
B.	high	< 1
C.	low	> 1
D.	low	< 1

26. Which of the following is **not** amphiprotic?

A. H_2O

B. HPO_4^{2-}

C. $H_2PO_4^-$

D. H_3O^+

27. The pH of a solution changes from 3 to 5. What happens to the concentration of hydrogen ions?

A. It increases by a factor of 2.

B. It increases by a factor of 100.

C. It decreases by a factor of 2.

D. It decreases by a factor of 100.

28. Which statement is correct about a Lewis base?

A. It is an electron pair donor and can act as a nucleophile.

B. It is an electron pair acceptor and can act as a nucleophile.

C. It is an electron pair donor and can act as an electrophile.

D. It is an electron pair acceptor and can act as an electrophile.

29. Which mixture forms a buffer solution with a pH < 7?

A. $50\,cm^3$ $0.10\,mol\,dm^{-3}$ $NH_4Cl\,(aq)$ + $50\,cm^3$ $0.10\,mol\,dm^{-3}$ $NH_3(aq)$

B. $50\,cm^3$ $0.10\,mol\,dm^{-3}$ $HCl\,(aq)$ + $100\,cm^3$ $0.10\,mol\,dm^{-3}$ $NH_3(aq)$

C. $50\,cm^3$ $0.10\,mol\,dm^{-3}$ $NaOH\,(aq)$ + $100\,cm^3$ $0.10\,mol\,dm^{-3}$ $CH_3COOH\,(aq)$

D. $50\,cm^3$ $0.10\,mol\,dm^{-3}$ $H_2SO_4(aq)$ + $100\,cm^3$ $0.10\,mol\,dm^{-3}$ $NH_3(aq)$

30. The equations below represent reactions involved in the Winkler method for determining the concentration of dissolved oxygen in water:

$$2Mn(OH)_2(s) + O_2(aq) \rightarrow 2MnO(OH)_2(s)$$
$$MnO(OH)_2(s) + 2H_2SO_4(aq) \rightarrow Mn(SO_4)_2(s) + 3H_2O(l)$$
$$Mn(SO_4)_2(s) + 2I^-(aq) \rightarrow Mn^{2+}(aq) + I_2(aq) + 2SO_4^{2-}(aq)$$
$$2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

What is the amount, in mol, of thiosulfate ions, $S_2O_3^{2-}(aq)$, needed to react with the iodine, $I_2(aq)$, formed by 1.00 mol of dissolved oxygen?

A. 2.00

B. 3.00

C. 4.00

D. 6.00

31. What are the products when molten sodium chloride is electrolysed?

	Cathode	**Anode**
A.	hydrogen	chlorine
B.	sodium	chloride
C.	sodium	chlorine
D.	chlorine	sodium

32. $E^{\ominus}$ values for some half-equations are given below.

$$Mn^{2+}(aq) + 2e^- \rightleftharpoons Mn(s) \qquad E^{\ominus} = -1.18\,V$$
$$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s) \qquad E^{\ominus} = -0.45\,V$$
$$Pb^{2+}(aq) + 2e^- \rightleftharpoons Pb(s) \qquad E^{\ominus} = -0.13\,V$$

Which reaction is spontaneous under standard conditions?

A. $Fe^{2+}(aq) + Mn(s) \rightarrow Fe(s) + Mn^{2+}(aq)$

B. $Mn^{2+}(aq) + Pb(s) \rightarrow Mn(s) + Pb^{2+}(aq)$

C. $Fe^{2+}(aq) + Pb(s) \rightarrow Fe(s) + Pb^{2+}(aq)$

D. $Mn^{2+}(aq) + Fe(s) \rightarrow Mn(s) + Fe^{2+}(aq)$

33. $50.0\,cm^3$ of $0.50\,mol\,dm^{-3}$ aqueous copper(II) sulfate, $CuSO_4(aq)$, is electrolysed using a current of 0.50 A for 30 minutes. What mass of copper, in g, is deposited on the cathode?
(M(Cu) = $64\,g\,mol^{-1}$; Faraday's constant (F) = $96500\,C\,mol^{-1}$.)

A. $\dfrac{50.0 \times 0.50 \times 64}{1000}$

B. $\dfrac{0.50 \times 30 \times 64}{96500 \times 2}$

C. $\dfrac{0.50 \times 30 \times 60 \times 64}{96500 \times 2}$

D. $\dfrac{50.0 \times 0.50 \times 64}{1000 \times 2}$

34. Which is propyl propanoate?

A. $CH_3CH_2CH_2OOCCH_2CH_3$

B. $CH_3CH_2CH_2COOCH_2CH_3$

C. $CH_3CH_2CH_2COCH_2CH_3$

D. $CH_3CH_2CH_2OCH_2CH_2CH_3$

35. Which could form an addition polymer?

A. $H_2NCH_2CHCHCH_2NH_2$

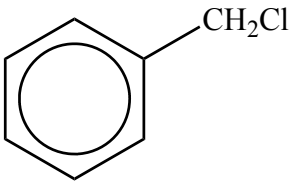
B. $H_2N(CH_2)_6CO_2H$

C. $HO(CH_2)_2CO_2H$

D. $H_2N(CH_2)_6NH_2$

36. Which benzene derivative can be formed from methylbenzene by electrophilic substitution?

A.



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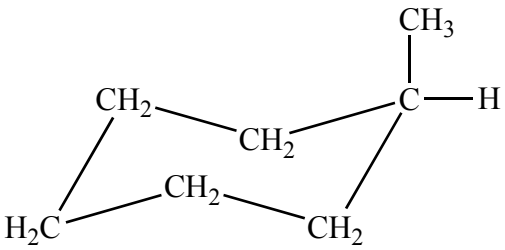
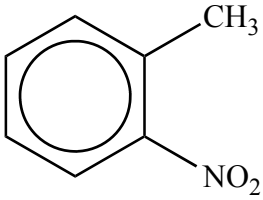
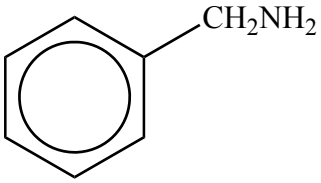


C.



D.



37. Which compound has two enantiomeric forms?

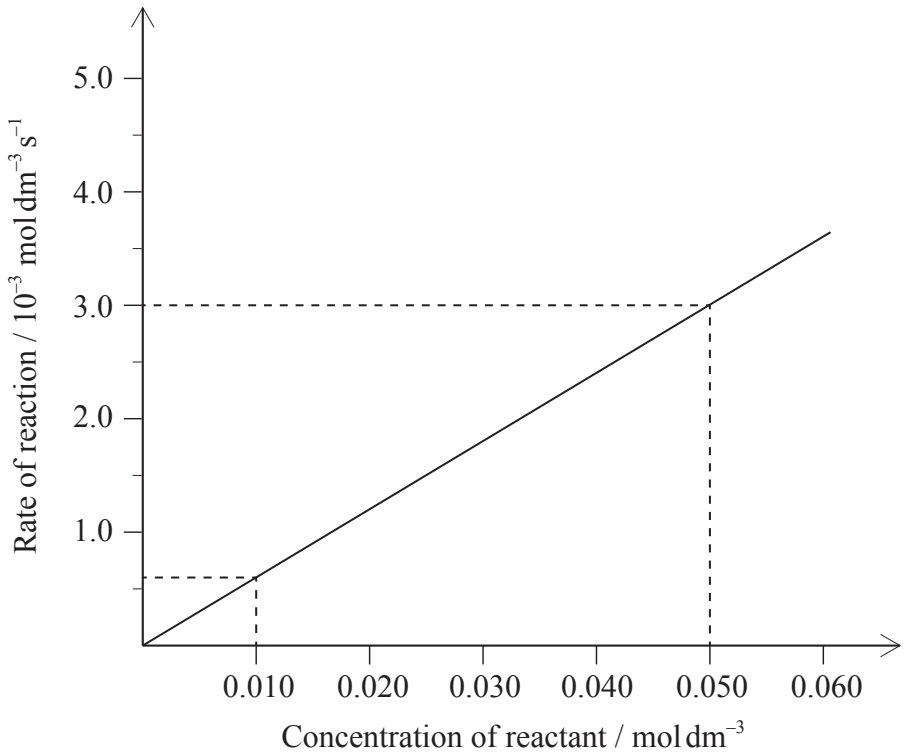
A. $CH_3CH_2CBr_2CH_3$

B. $CH_3CH_2CHBrCH_3$

C. $CH_3(CH_2)_2CH_2Br$

D. $CH_3CH_2CHBrCH_2CH_3$

38. Which combination in the table correctly states the value and units of the gradient?



	Value	Units
A.	$\frac{3.0\times10^{-3}-0.6\times10^{-3}}{0.050-0.010}$	s^{-1}
B.	$\frac{3.0\times10^{-3}-0.6\times10^{-3}}{0.050-0.010}$	s
C.	$\frac{0.050-0.010}{3.0\times10^{-3}-0.6\times10^{-3}}$	s^{-1}
D.	$\frac{0.050-0.010}{3.0\times10^{-3}-0.6\times10^{-3}}$	s

39. Which technique involves the absorption of radiation by bonds between atoms?

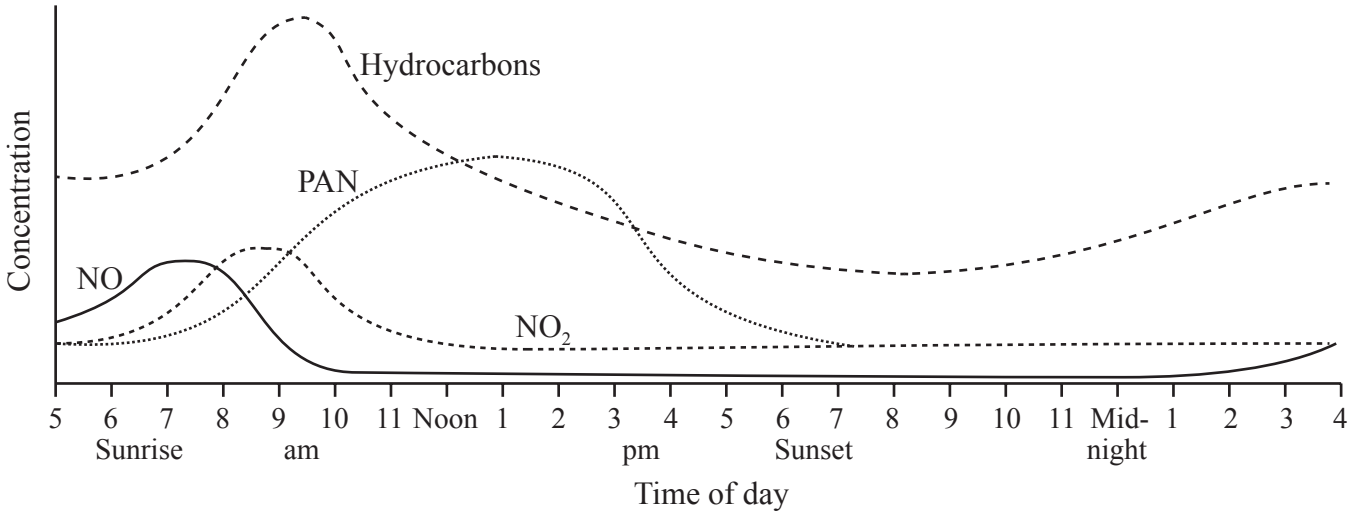
A. ^{1}H NMR

B. Infrared spectroscopy

C. X-ray crystallography

D. Mass spectrometry

40. The graph shows the concentration of some pollutants in a city over a 24-hour period.



Which of the following could **not** be inferred from the graph?

A. Hydrocarbons cause less harm to health than PAN.

B. An increase in hydrocarbons is caused by the morning rush hour.

C. PAN concentration increases as the intensity of sunlight increases.

D. NO_2 production follows the production of NO.